Automated UI Test Suite – Final Assignment

Python · Pytest · Selenium WebDriver · Chrome

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Project Overview

- **Application Under Test:** Photo Tech World – a camera & gear portal
- **Source files:** index.html (UI structure), script.js (validation logic), style.css
- **Login credentials:** username: **admin** / password: **@Dm1n**
- **3 navigation tabs:** Cameras · Lenses · Contact
- **Contact form fields:** Name, Email, Phone, Message

The test suite covers **21 individual test cases** across 5 test classes.

Technology Stack

Layer	Tool	Role
Language	Python 3.x	Implementation language
Test Framework	pytest	Test runner, fixtures, assertions
UI Automation	Selenium WebDriver	Browser control & DOM interaction
Browser	Google Chrome	Mandatory target browser
Driver Mgmt	ChromeDriver	Bridges Selenium and Chrome
Wait Strategy	WebDriverWait / EC	Explicit waits for dynamic elements

Test Classes & Coverage

Class	Tests	What is verified
TestLogin	5	Login page, invalid creds, valid login, initials
TestNavigation	5	Tab switching, single-page visibility
TestContactForm	8	Name / Email / Phone validation rules
TestSendMessage	2	Success & failure of Send Message button
TestLogout	1	Logout button returns to login page

Key Implementation Details

Fixture Design

• Module-scoped driver (one Chrome instance per run) • autouse fixtures navigate to the correct page • login() helper reused across multiple classes

Locator Strategy

- By.ID for inputs and error divs (stable)
- By.XPATH for buttons without IDs
- execute_script() for inline JS validators

Known Bug in source

Inscript.js, submitForm() tries to add an event listener to #myForm which does not exist. This causes a TypeError in the console but does not break the success message. Tests are written to avoid this path.

How to run

```
pip install pytest selenium
```

```
pytest st2521041_py_test.py -v
```

Conclusions & Deliverables

- 21 automated tests covering all requirements from the assignment brief
- Tests structured into clear classes by feature area
- Headless Chrome – runs in CI/CD environments without a display
- One known source-code bug identified and documented

File	Description
st2521041_py_test.py	Full Selenium / pytest test suite
st2521041_py_doc.pdf	This documentation
st2521041_py_present.pdf	Presentation for oral defence